Siwar M'SALLEM

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with hands-on experience developing secure web and mobile applications using React, React Native, TypeScript, Node.js, and Fastify. Experienced in RESTful API development, multi-provider payment integrations (Tap, Tamara, Tabby), and secure transaction workflows. Skilled in MongoDB, microservices, AI-powered features, and CI/CD automation with Docker and GitHub Actions.

TECHNICAL SKILLS

Frontend & Mobile: React.js, React Native (Expo), Next.js, TypeScript, Redux Toolkit, Tailwind CSS, Material UI, Vite

Backend & APIs: Node.js, Fastify, Express.js, REST APIs, Socket.IO, FastAPI, Flask, Laravel, Spring Boot

Programming Languages: JavaScript, TypeScript, Python, Java, SQL

Databases: MongoDB, Mongoose, MySQL, SQL Server

Security & Payment Integration: JWT, RBAC, Zod, Webhooks, Tap Payments, Tamara, Tabby, Stripe

Testing & API Tools: Vitest, MongoDB Memory Server, JUnit, Mockito, Postman, Swagger/OpenAPI

DevOps & Version Control: Docker, Docker Compose, GitHub Actions, Jenkins, Git, GitHub, SonarQube

PROFESSIONAL EXPERIENCE

Software Engineer – Kochi SaaS Platform Aug 2026 – Oct 2026
Tradrly *Mahdia, Tunisia*

- Developed full-stack features for an Arabic-first fitness coaching SaaS using Fastify, TypeScript, MongoDB, and React Native/Expo, supporting client and coach workflows.
- Integrated multiple payment providers (Tap, Tamara, Tabby) and Apple Pay via Tap, implementing checkout, payment verification, capture, and refund workflows.
- Enhanced transaction security and reliability through webhook validation, payment state management, idempotency safeguards, and duplicate-capture prevention.
- Implemented multi-currency pricing with SAR conversion, payment-time price snapshots, and configurable platform fees and taxes.
- Migrated AI-powered features to Anthropic Claude, integrating chat, content generation, and OCR services with robust error handling, timeouts, and retries.

Full Stack Software Engineer Intern – Engineering Final Year Project Feb 2026 – Jul 2026
Tradrly *Mahdia, Tunisia*

- Designed and developed a MERN-based e-learning platform with dedicated dashboards for students, teachers, and administrators.
- Architected RESTful APIs using Node.js, Express.js, and MongoDB for course management, enrollments, learning resources, and virtual classrooms.
- Built responsive React.js interfaces using Vite and Tailwind CSS, delivering role-specific dashboards and interactive learning workflows.
- Implemented secure JWT authentication and RBAC, and integrated FastAPI microservices for AI-powered quiz generation, lesson assistance, and semantic search.
- Containerized the application using Docker and Docker Compose and implemented CI/CD pipelines with GitHub Actions.

Full Stack Software Engineer Intern Jul 2025 – Aug 2025
Tradrly *Mahdia, Tunisia*

- Developed an AI-powered video dubbing platform using React.js, Node.js, Express.js, and MongoDB, enabling automated multilingual video localization.
- Built responsive React.js interfaces with Vite, Tailwind CSS, and Material UI, supporting multilingual LTR/RTL layouts.
- Designed RESTful APIs for authentication, video uploads, and processing workflows, integrating Flask-based AI services with Whisper, Gemini, and FFmpeg.
- Implemented JWT authentication, secure file management, and asynchronous communication between Express and AI processing services.

Web Development Intern Jul 2024 – Aug 2024
Telnet *Tunis, Tunisia*

- Contributed to Pulse HR, an enterprise HR management platform, developing reusable interfaces using Next.js, React, TypeScript, and Material UI.
- Developed and integrated RESTful APIs using Laravel, Eloquent ORM, and MySQL for employee management, recruitment, and organizational workflows.
- Implemented client-side state management with Redux Toolkit and React Query, collaborating in Agile development, debugging, and testing.

Bachelor's Final Year Project – Full Stack Development Feb 2023 – Jun 2023
StartNow *Tunis, Tunisia*

- Contributed to a full-stack business application using React.js, Redux, Express.js, MongoDB, and SQL Server.
- Developed reusable React components and configurable AG Grid tables with advanced filtering, data visualization, and customization features.
- Built and integrated RESTful APIs with Express.js, supporting business workflows and database operations in an Agile development environment.

PROJECTS

Khedma – Service Marketplace Platform | *React, React Native, Fastify, MongoDB* Aug 2026 – Present

- Developed a full-stack service marketplace with React/TypeScript web and React Native/Expo mobile applications, backed by a modular Fastify API and MongoDB.
- Implemented secure authentication with JWT access and refresh tokens, email OTP verification, role-based access control, and professional onboarding workflows.
- Built an end-to-end service request workflow using MongoDB transactions to atomically accept proposals and prevent duplicate professional assignments.
- Integrated real-time notifications with Socket.IO and secure messaging with Stream Chat, enforcing access controls and protecting sensitive user data.

EcoEvents – Ecological Event Management Platform | *Laravel 12, MySQL, Docker* Oct 2025 – Dec 2025

- Developed a full-stack event management platform using Laravel and MySQL, supporting event organization, participant registration, partnerships, and digital resource management.
- Implemented secure authentication and role-based access with Laravel Breeze and Jetstream, and built responsive interfaces using Blade and Tailwind CSS.
- Containerized the application with Docker and integrated SonarQube, Prometheus, and Grafana for code quality analysis and monitoring.

TheMenuFy – AI-Powered Meal Ordering Platform | *React, Node.js, Flask* Feb 2025 – May 2025

- Developed a full-stack meal ordering platform using React, Express.js, and Flask, integrating an AI-powered chatbot and personalized meal recommendation system.
- Integrated Stripe payment processing and external REST APIs to support secure payment and ordering workflows.
- Containerized the application using Docker and implemented a CI/CD workflow with SonarQube for automated code quality analysis.

EDUCATION

Engineering Degree in Computer Engineering Sep. 2026
ESPRIT – Engineering School *Tunisia*

Bachelor's Degree in Computer Science 2020 – 2023
Higher Institute of Computer Science and Mathematics of Monastir (ISIMM) *Monastir, Tunisia*

- Specialization: Software Engineering and Information Systems.

LANGUAGES

- **Arabic:** Native
- **French:** Professional Working Proficiency
- **English:** Professional Working Proficiency
- **Italian:** Beginner (A2)